

Missionary Memorabilia Analysis

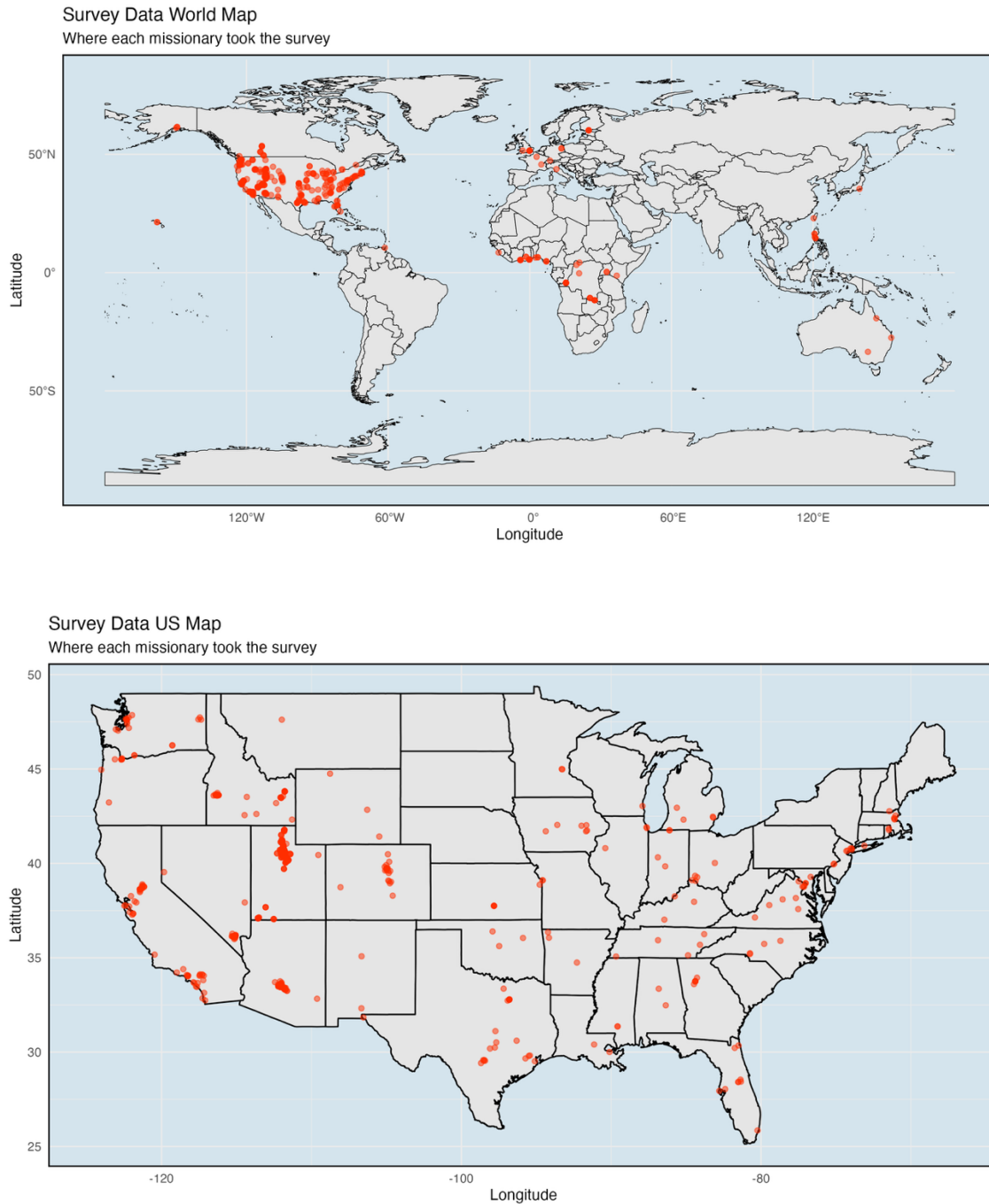
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1.1 Survey Taker Geographic Breakdown



As far as I can tell, these locations are as accurate as Qualtrics can make them and I have no further way of proving if there are more people responding in areas such as Central and South America. We can also see from these graphs that as expected, most of the people taking the survey are in the U.S.

1.2 Demographic Breakdown

	Overall (N=1231)
Mission Region	
U.S./Canada	455 (37.0%)
Europe	286 (23.2%)
South America	154 (12.5%)
Africa	98 (8.0%)
Asia	70 (5.7%)
Central America	60 (4.9%)
Oceania	36 (2.9%)
Multiple Regions	67 (5.4%)
Unknown	5 (0.4%)
Missionary	
Elder	759 (61.7%)
Sister	472 (38.3%)
Time Since Mission	
Less than 6 months	93 (7.6%)
6-12 months	161 (13.1%)
1-3 years	265 (21.5%)
3-5 years	155 (12.6%)
5-10 years	222 (18.0%)
10-20 years	65 (5.3%)
20-30 years	90 (7.3%)
30-40 years	54 (4.4%)
40+ years	126 (10.2%)
Memorabilia Importance	
Not at all important	14 (1.1%)
Slightly important	84 (6.8%)
Moderately important	189 (15.4%)
Very important	366 (29.7%)
Extremely important	378 (30.7%)
Unknown	200 (16.2%)

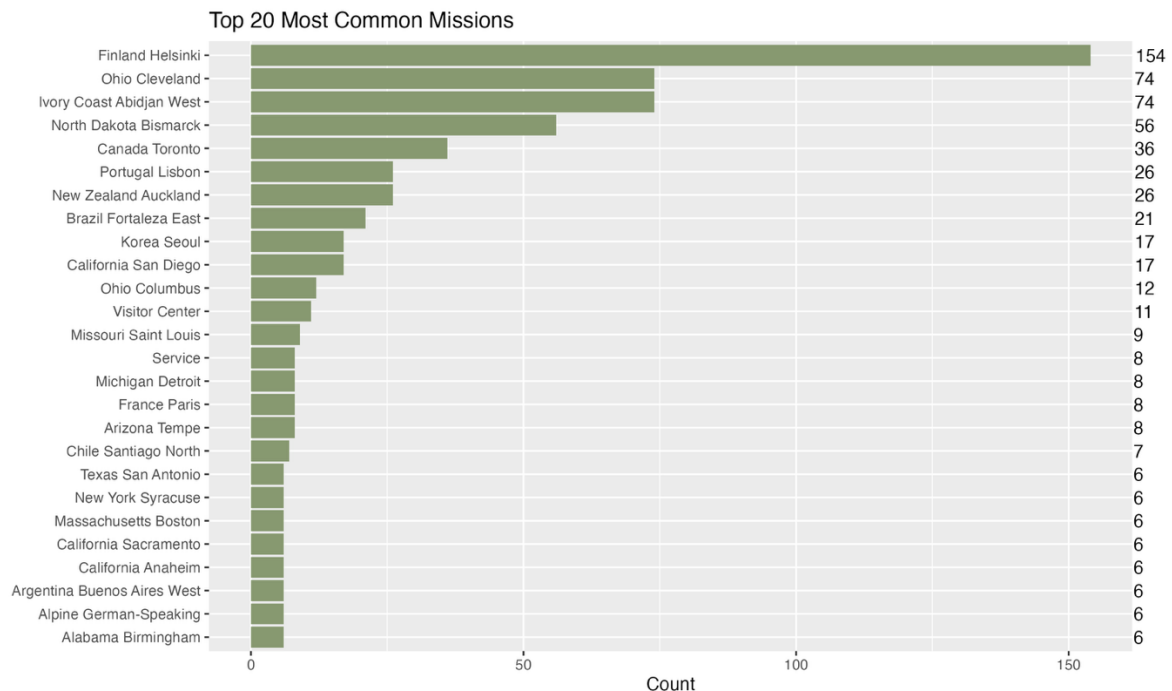
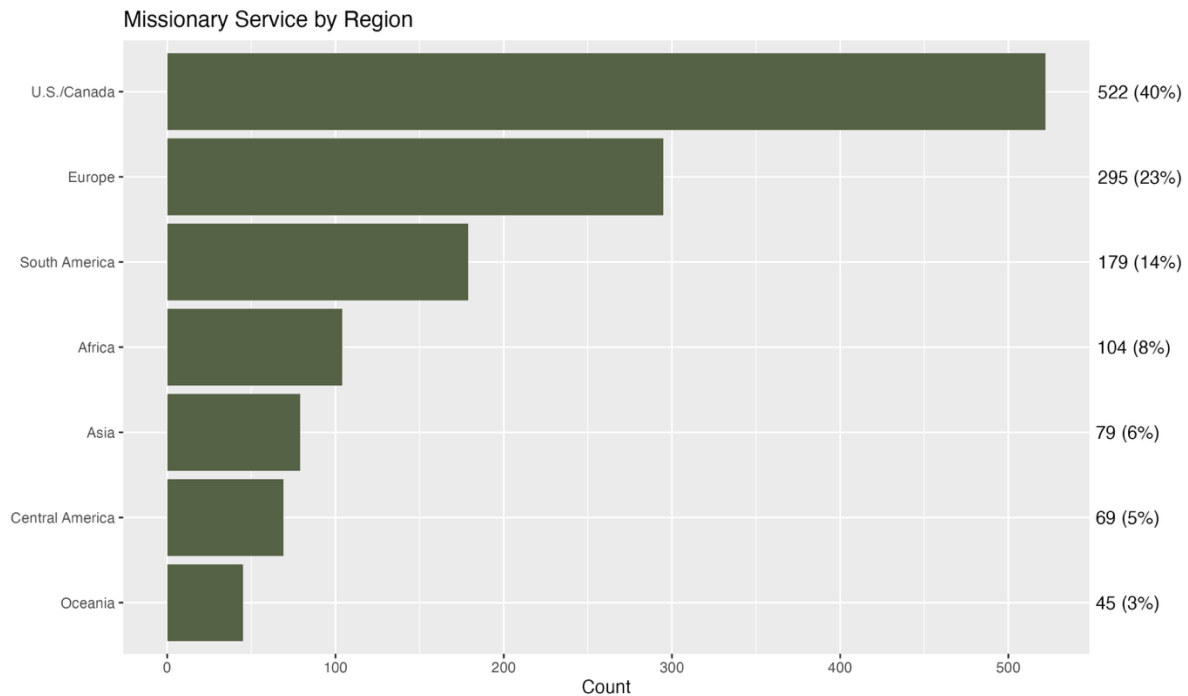
This is a breakdown of the survey takers by where they served their mission, if they were an Elder or Sister missionary, how long it's been since they served their mission, and how important the memorabilia was to themselves. The most important takeaway I see is the fact that we don't have many people who didn't value their missionary memorabilia.

In our survey we had a total of 1,308 respondents. Of those respondents, 77 were excluded due to missing data (49 were completely blank, 27 answered neither the memorabilia items nor importance question, and 1 was missing time since missionary service). There were also 200 respondents which didn't answer the memorabilia importance question, but had filled out all previous questions. These people were included in the item counts, but excluded from the memorabilia importance analysis.

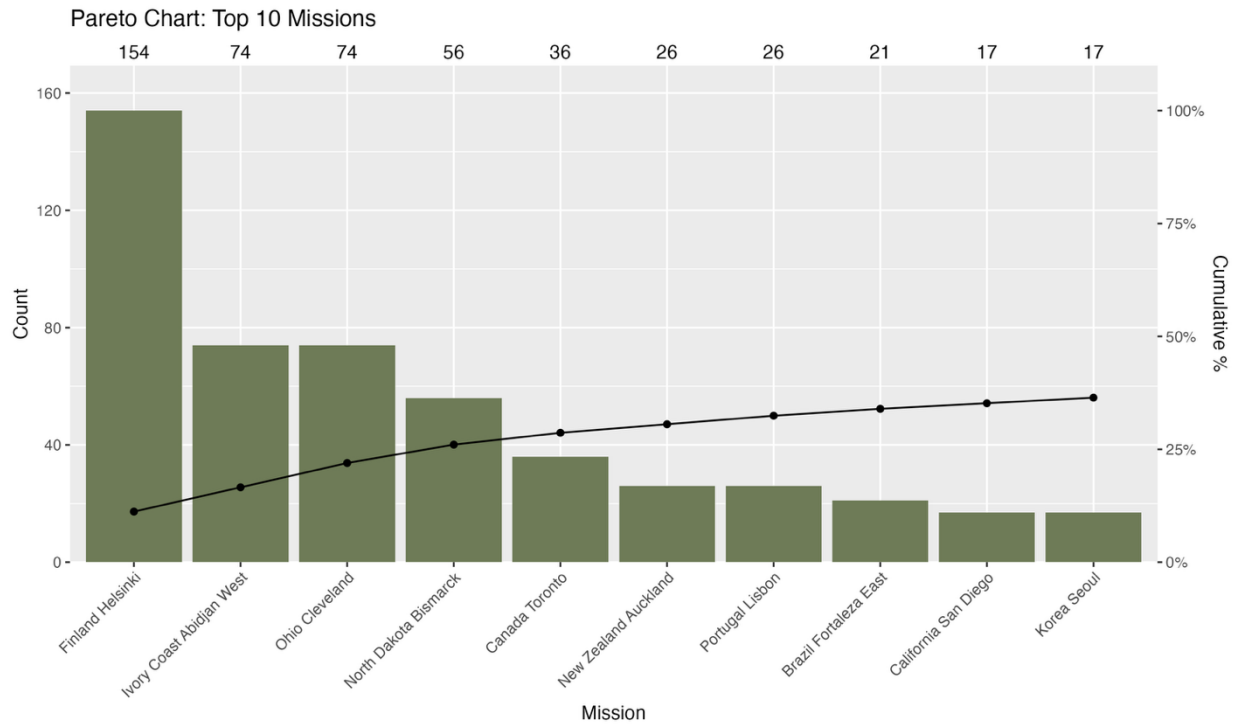
Region was defined as the following:

U.S./Canada	All U.S. states and Canadian provinces
Central America	All countries from Mexico to Panama including the Caribbean islands
South America	All countries in continental South America
Europe	All countries in continental Europe including all of Russia and European territories
Asia	All countries in continental Asia
Africa	All countries in continental Africa
Oceania	All countries including Australia, New Zealand, and the Pacific Islands
Multiple Regions	Respondents who served in more than one of the above regions

1.3 Missionary Service Regions



This shows the regions missionaries served in as well as the top 20 missions that were in the survey. I feel that Central America is underrepresented in our data and would've expected more people to be serving in that area.

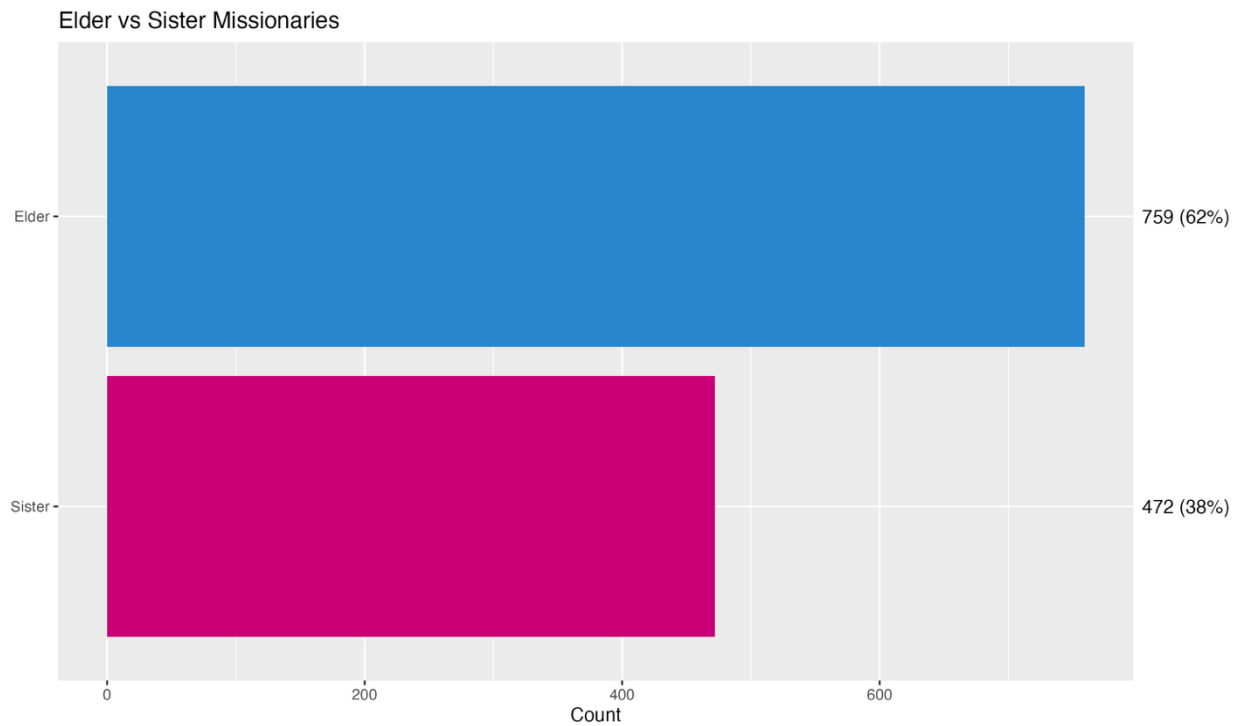


This is a Pareto Chart showing the top 10 missions and their cumulative percentage of the number of people who were in those missions. The top 10 missions only account for about a third of the people in our dataset, meaning we got a pretty good variety of missionaries from different missions in our dataset.

Number of Missionaries	Number of Missions
1	177
2	95
3	59
4	34
5	15
6	8
7	1
8	4
9	1
11	1
12	1
17	2
21	1
26	2
36	1
56	1
74	2
154	1
Total Number of Missions: 406	

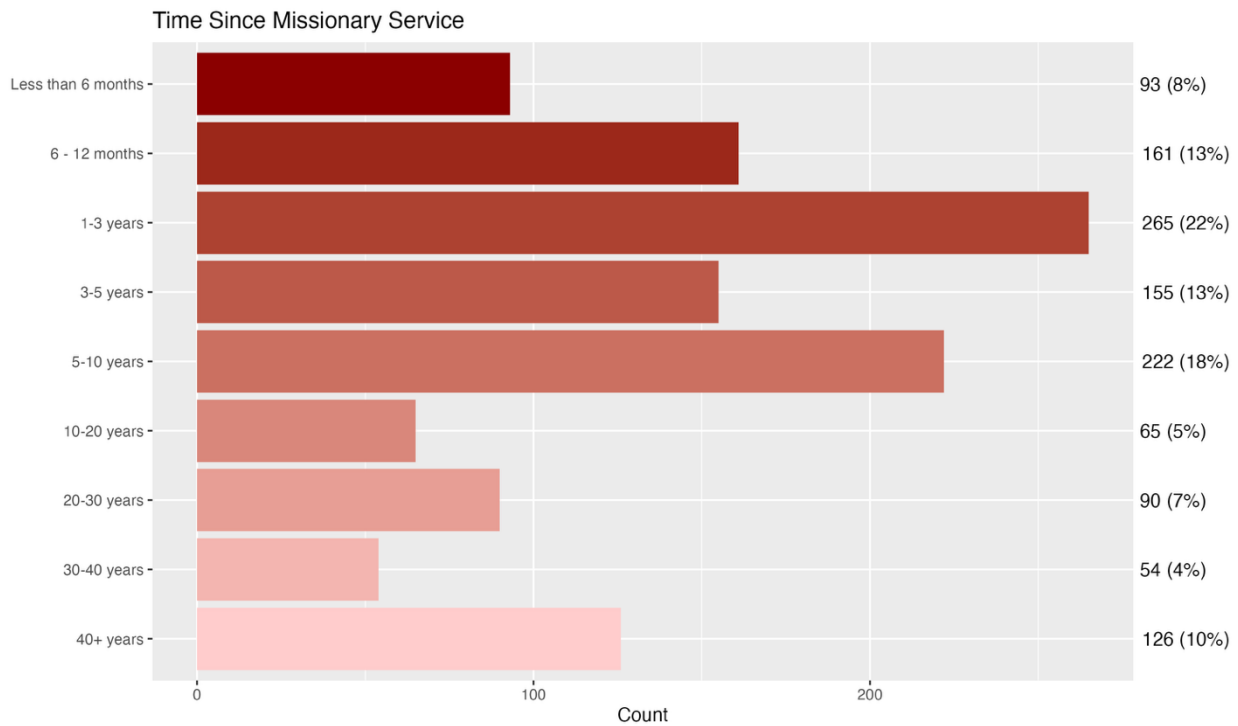
This table shows how many missions had the same number of missionaries report. As we can see, the distribution is right skewed with many missions only having one person in the mission and fewer people as the number of missionaries gets larger.

1.4 Elder and Sister Missionaries



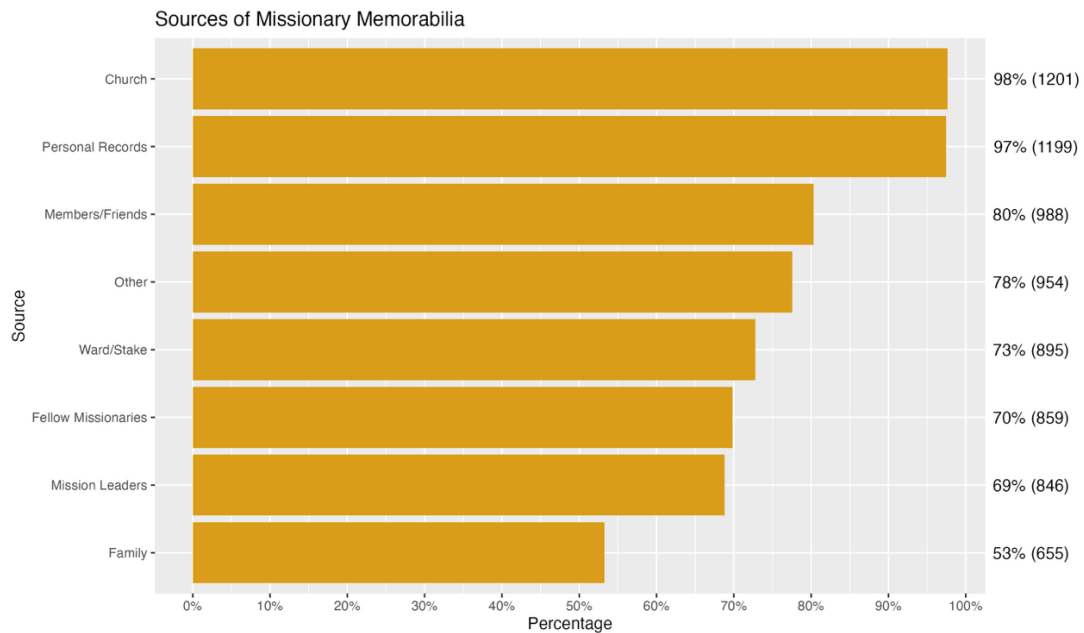
These graphs show the counts as well as the percentage of missionaries who were Elders and Sister Missionaries. I think that we got a pretty sizable amount of sister missionaries to respond to our survey given the amount I would've expected to respond to the survey.

1.5 Time Since Missionary Service

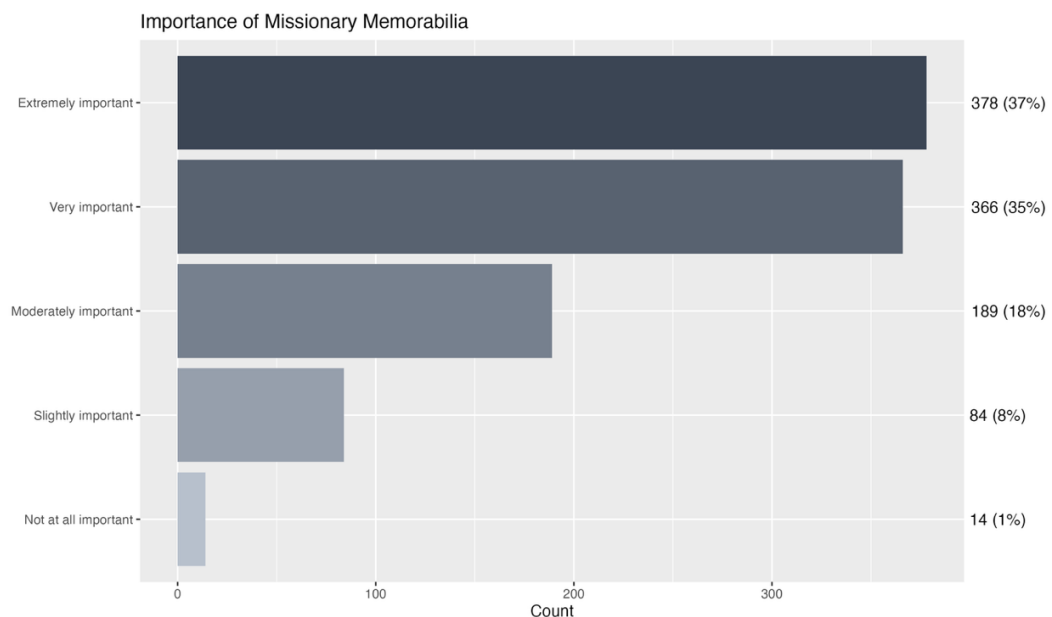


These graphs show the distribution of both counts and percentages of how long it's been since the missionary did their missionary service. I think that we got a pretty good distribution of people responding to the survey to represent all time periods.

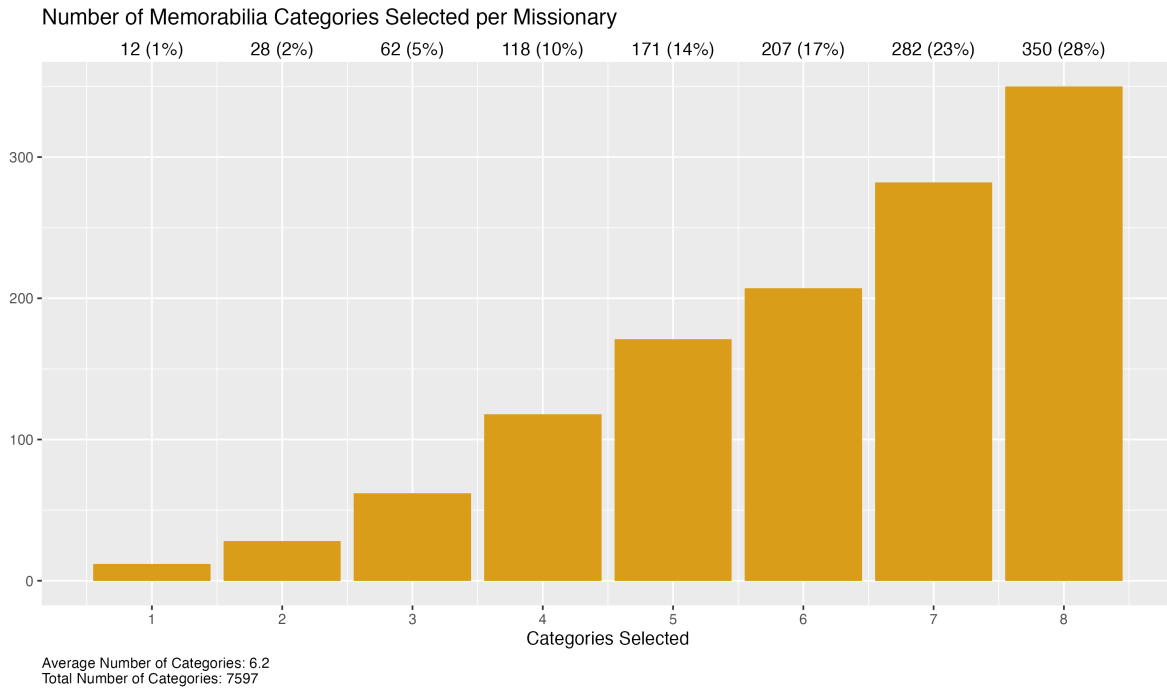
1.6 Memorabilia Categories and Importance



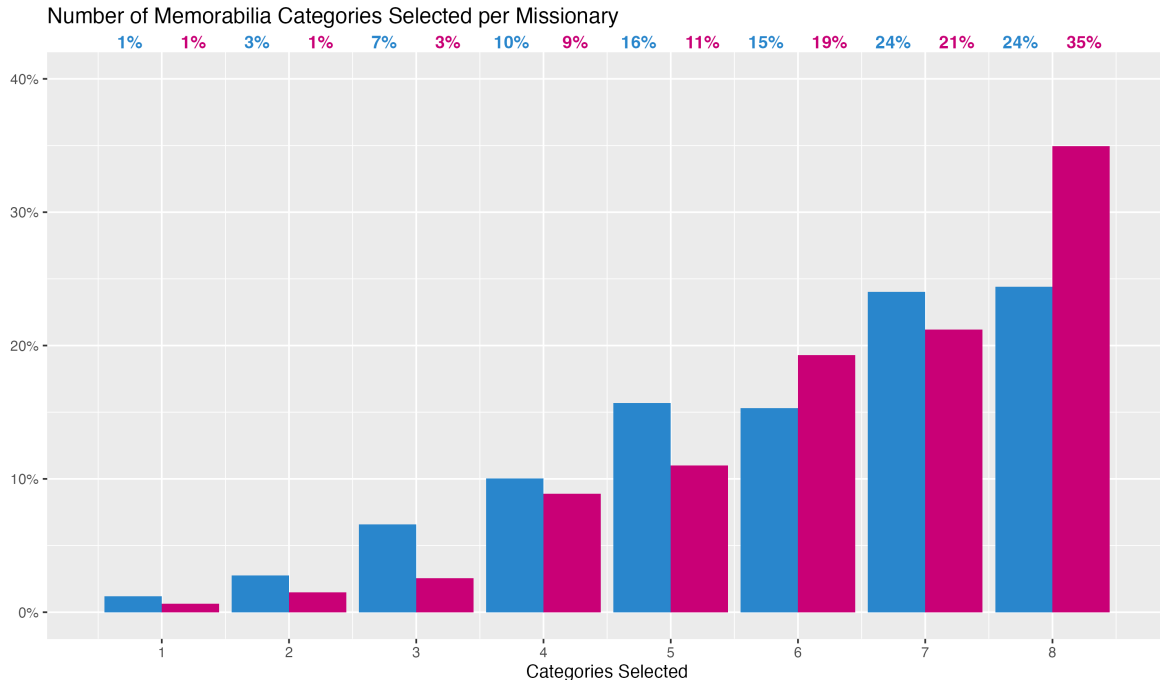
This graph shows the percentage of people who selected each of the memorabilia sources. You can tell that there are three distinct levels which people were selecting the memorabilia categories.



These graphs show the number of people as well as the percentage of people and how much importance they give to their missionary memorabilia. As mentioned before, we don't have a ton of people who didn't value their memorabilia.



This graph shows the number of memorabilia categories that each missionary selected. Lines up well given the fact that most missionaries were identifying with some of the categories.

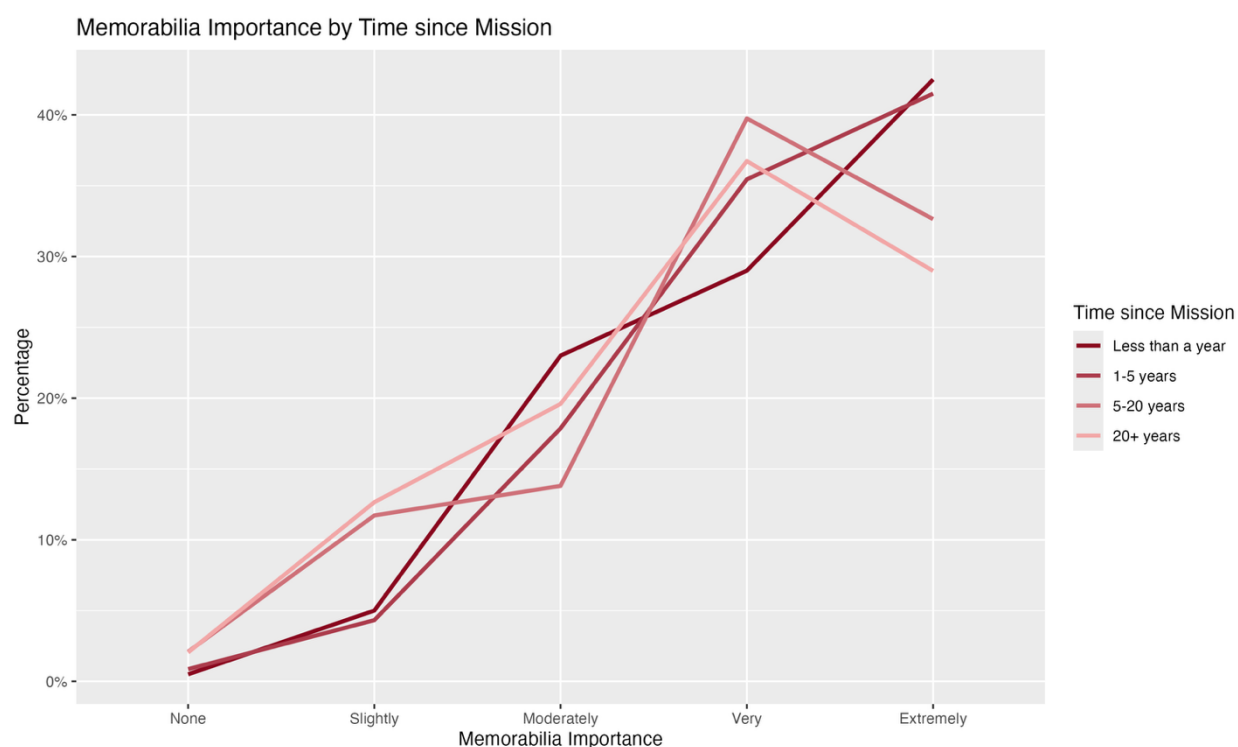


These graphs show the percentage of memorabilia categories that were selected by Elder and Sister missionary. We can see that Sisters tend to keep something from every category, while the Elders lean to keep more, but not quite as much.

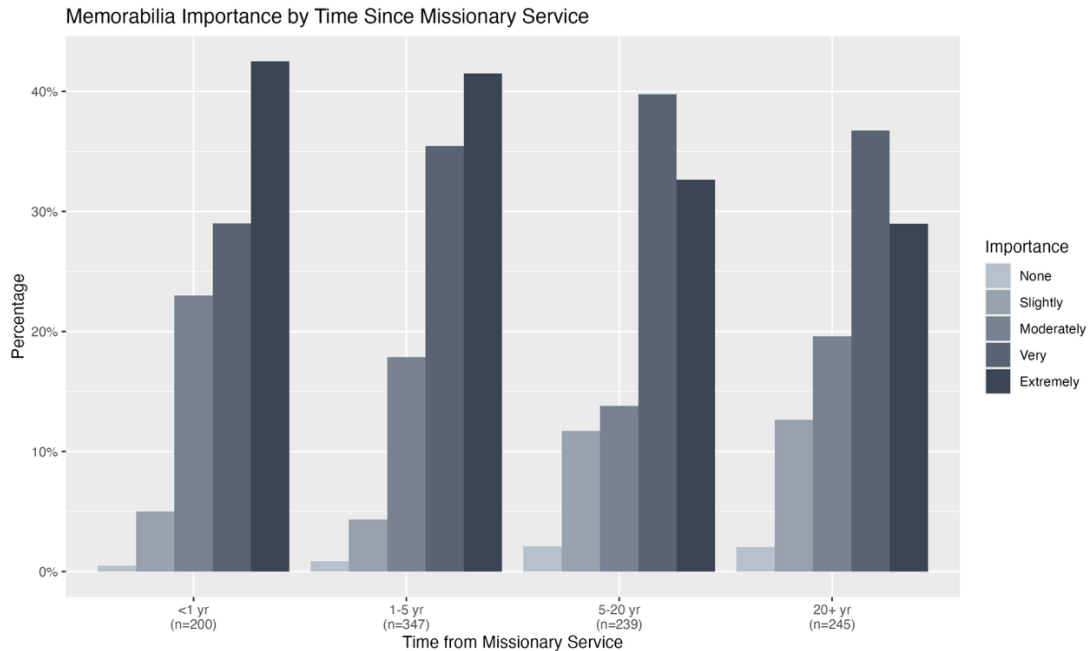
1.7 Time Since Mission and Memorabilia Importance

Time from Missionary Service	Not at all important	Slightly important	Moderately important	Very important	Extremely important	Total
<1 yr	0.5% (1)	5.0% (10)	23.0% (46)	29.0% (58)	42.5% (85)	100.0% (200)
1-5 yr	0.9% (3)	4.3% (15)	17.9% (62)	35.4% (123)	41.5% (144)	100.0% (347)
5-20 yr	2.1% (5)	11.7% (28)	13.8% (33)	39.7% (95)	32.6% (78)	100.0% (239)
20+ yr	2.0% (5)	12.7% (31)	19.6% (48)	36.7% (90)	29.0% (71)	100.0% (245)
Total	1.4% (14)	8.1% (84)	18.3% (189)	35.5% (366)	36.7% (378)	100.0% (1,031)

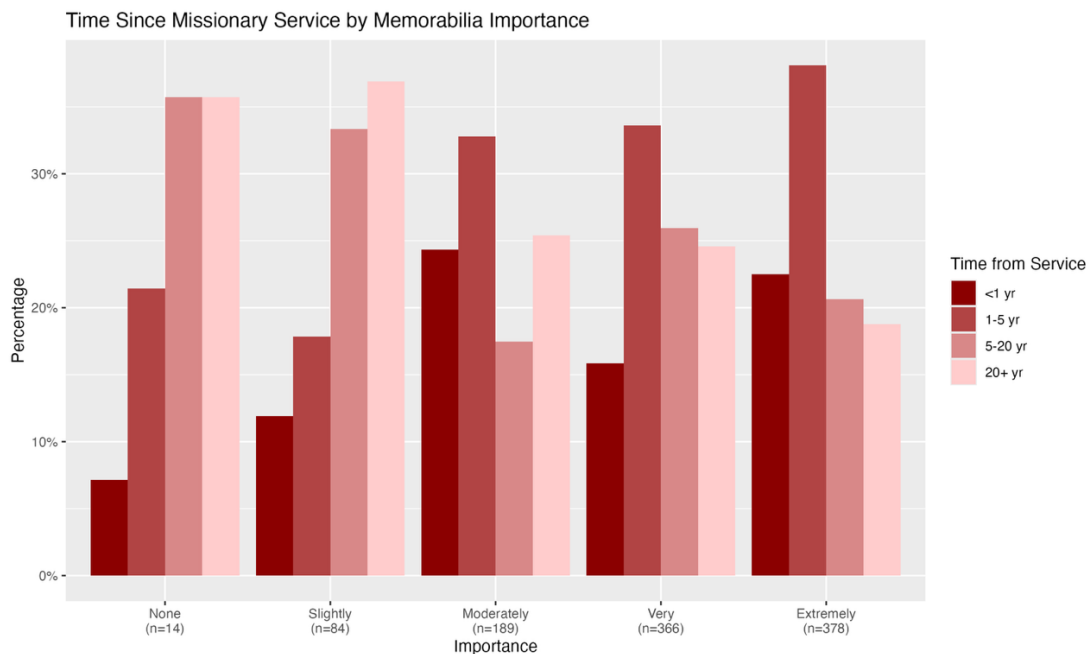
This is a cross table of time since missionary service and memorabilia importance. When performing a Monte Carlo chi-square test to see if these variables are independent of each other, it was found that there is an association between the two, $\chi^2(12, N = 1,031) = 40.07$, $p \approx 0.0002$.



This graph shows the percentage of memorabilia importance by time since missionary service. The graph starts with low memorabilia importance and goes to extremely high memorabilia importance. We can see that they all have an upwards trend, with the peak being around very important. However, at extremely important, the two more recent groups go up, while the two less recent groups go down.

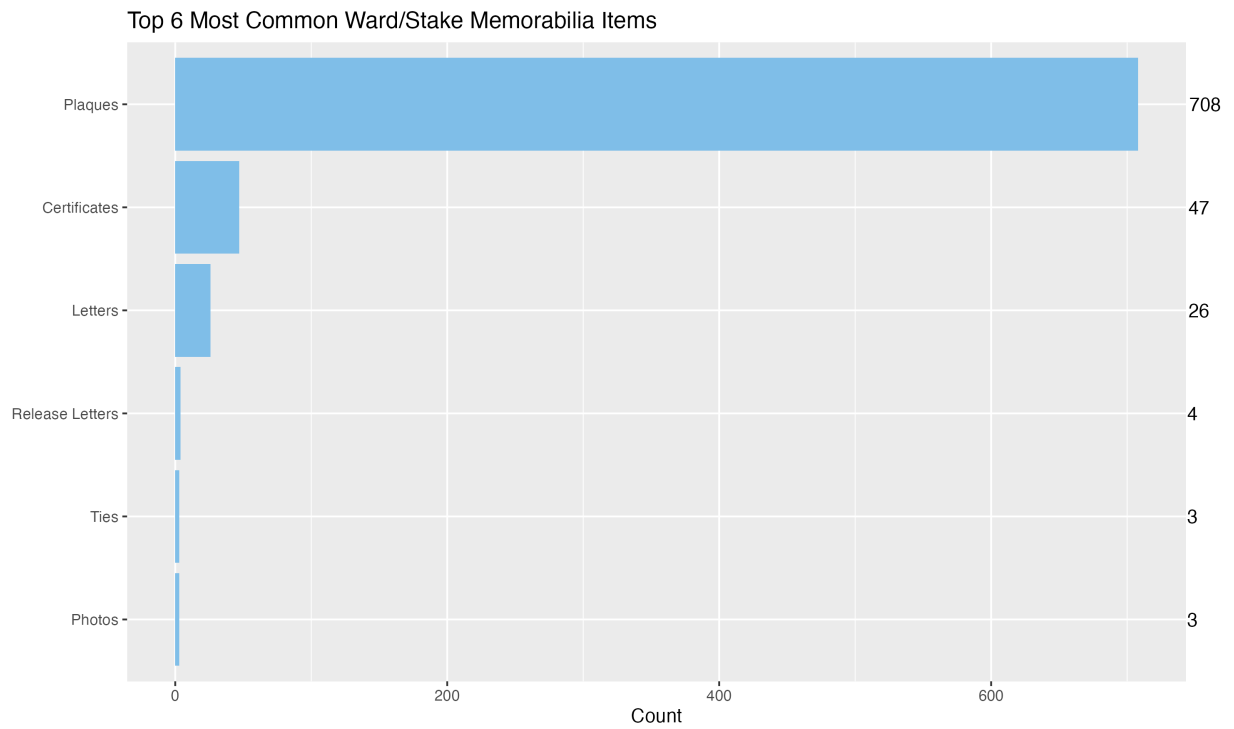
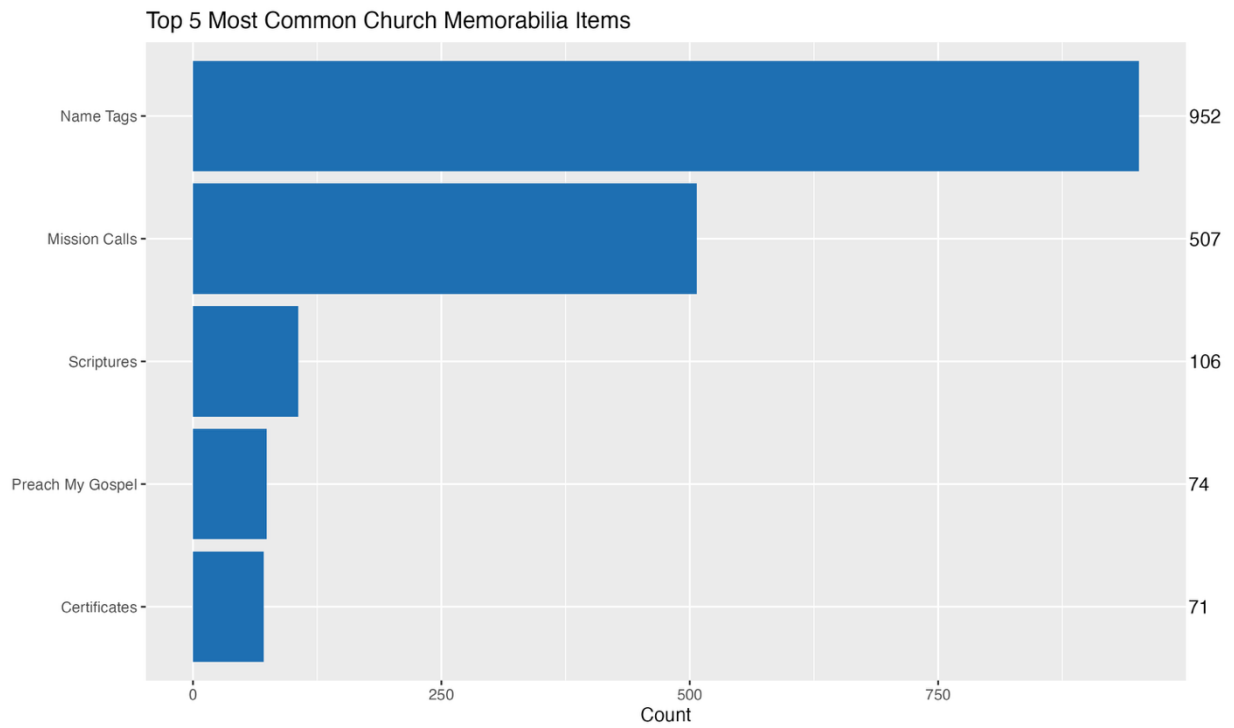


This is a bar chart of the percentages of memorabilia importance by time since missionary service. You can see a very small increase in no importance as the time goes on as well as a decrease in extreme importance as time goes on.

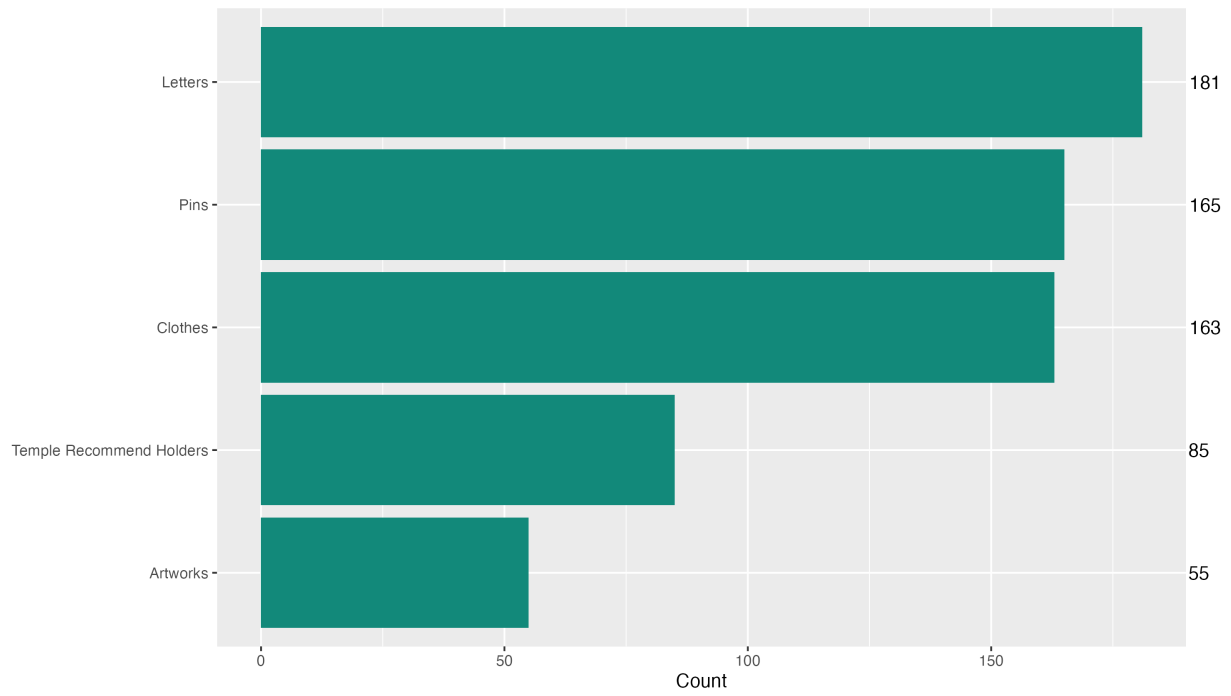


This is a bar chart of percentages of time since missionary service by memorabilia importance. You can see the trends growing for the more recent returns and diminishing for the later returns, although the sample size should be noted since low importance has a small sample size whereas high importance has a very large sample size.

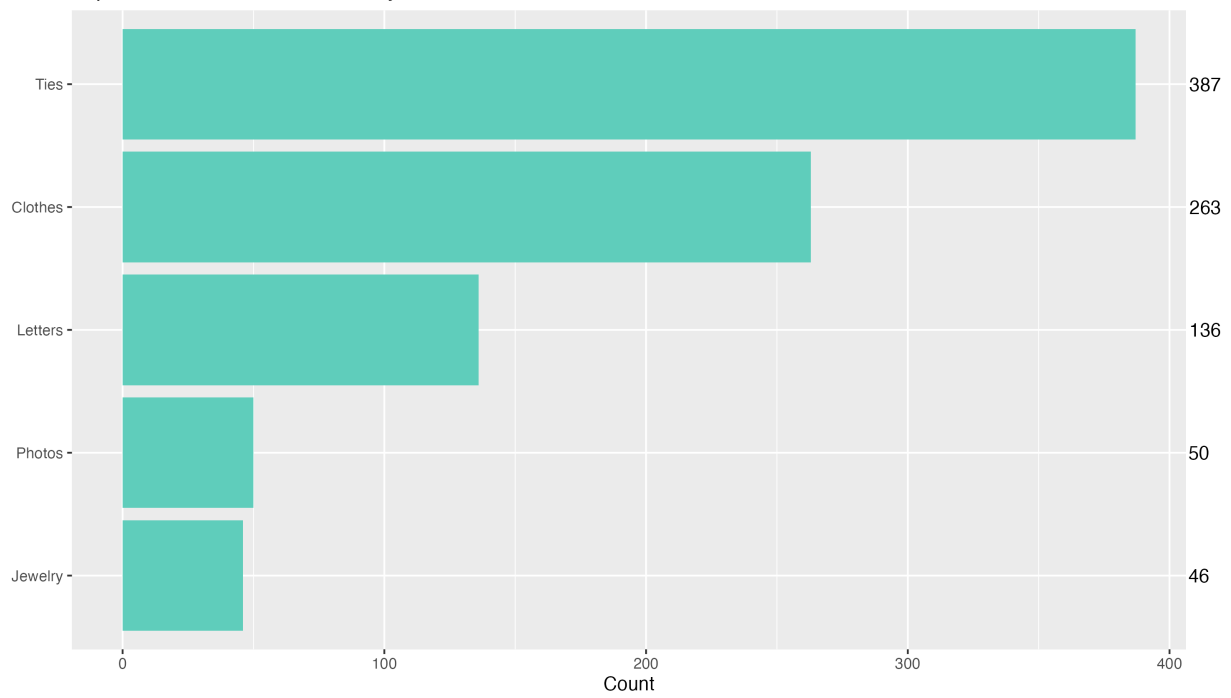
1.8 Top Memorabilia Items



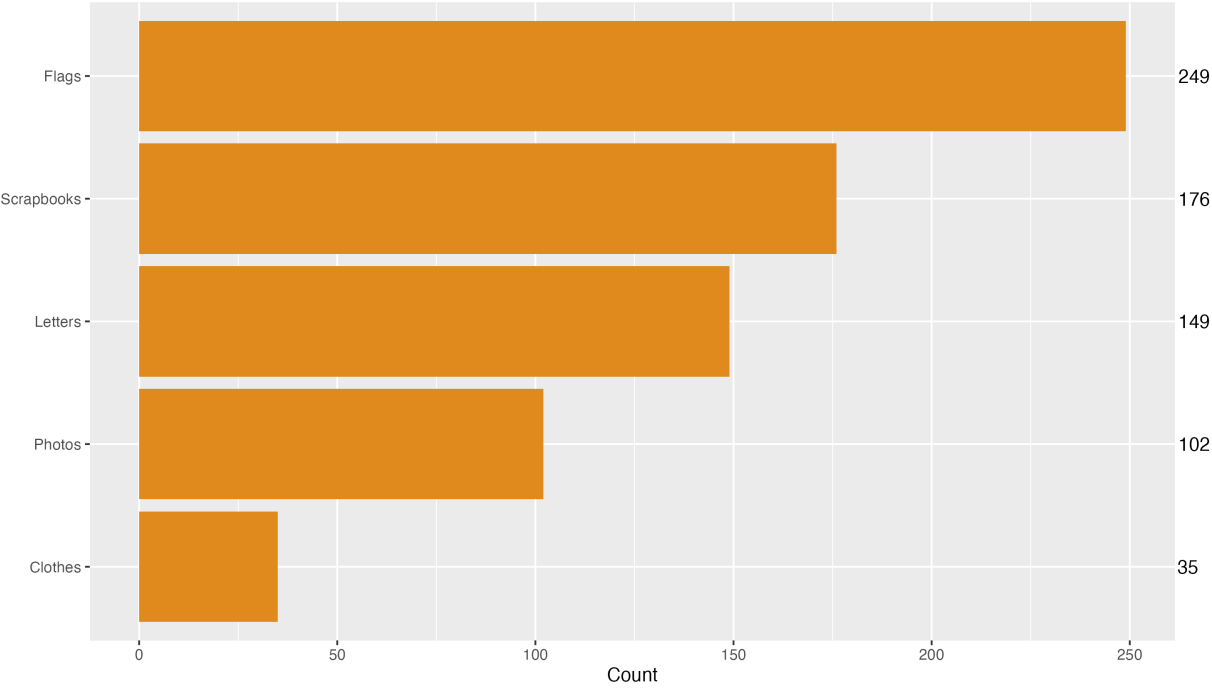
Top 5 Most Common Mission Leader Memorabilia Items



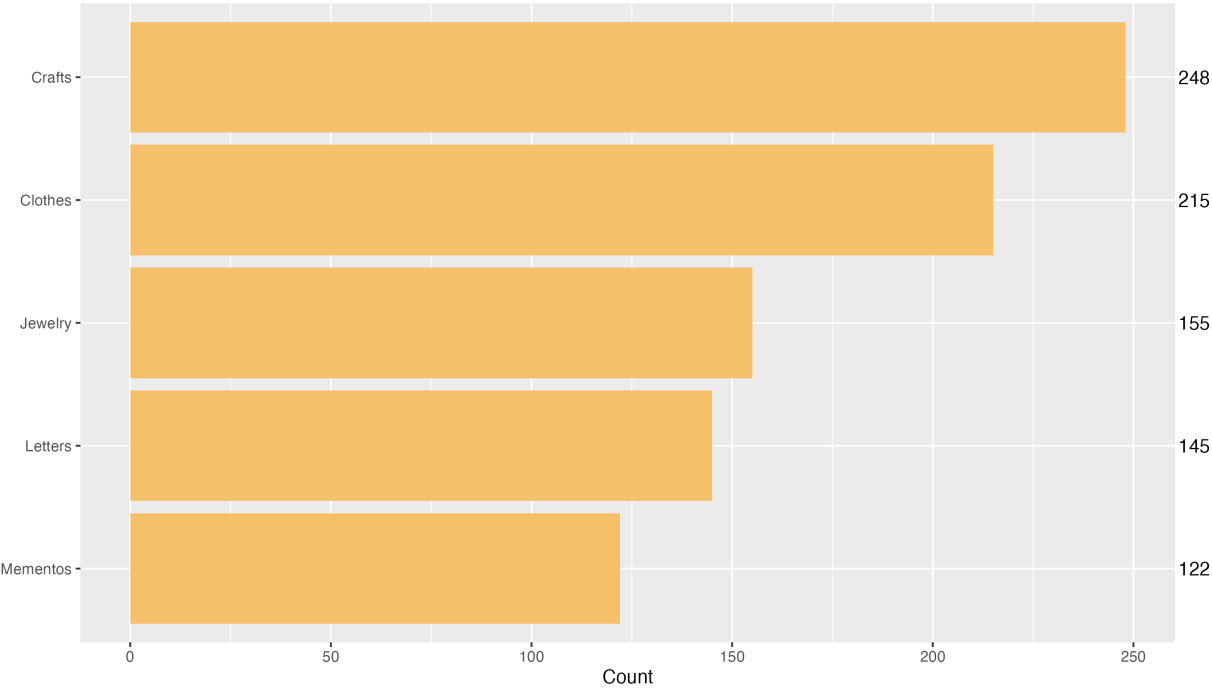
Top 5 Most Common Missionary Memorabilia Items



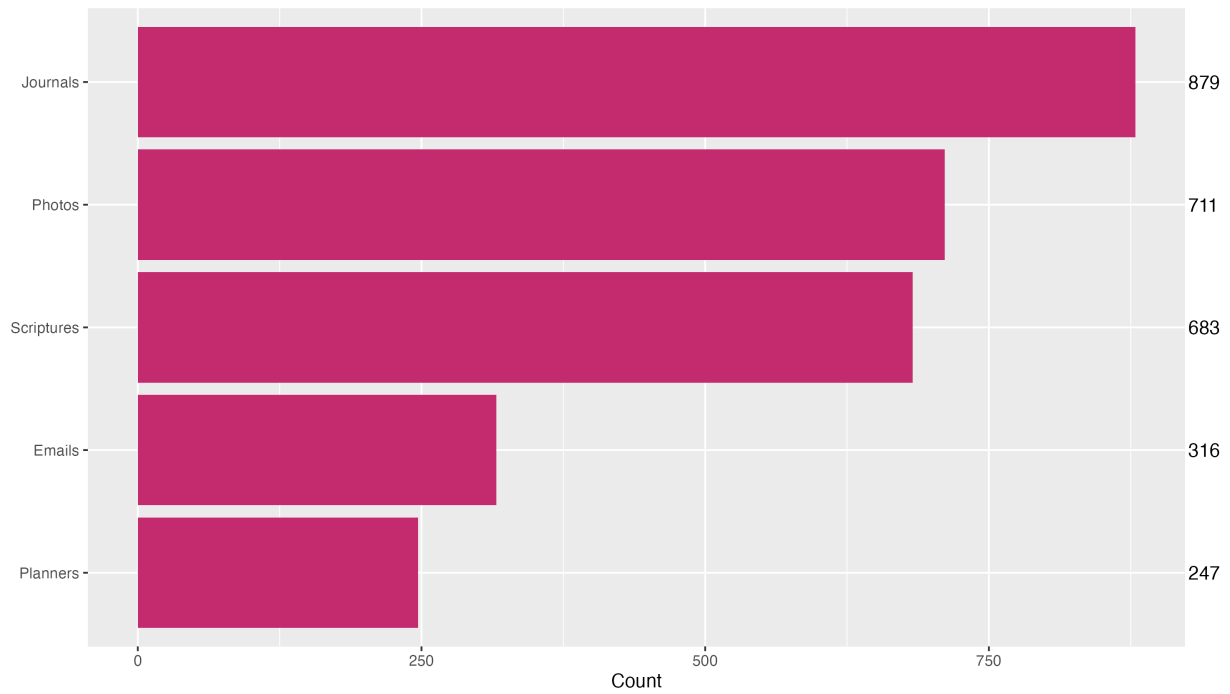
Top 5 Most Common Family Memorabilia Items



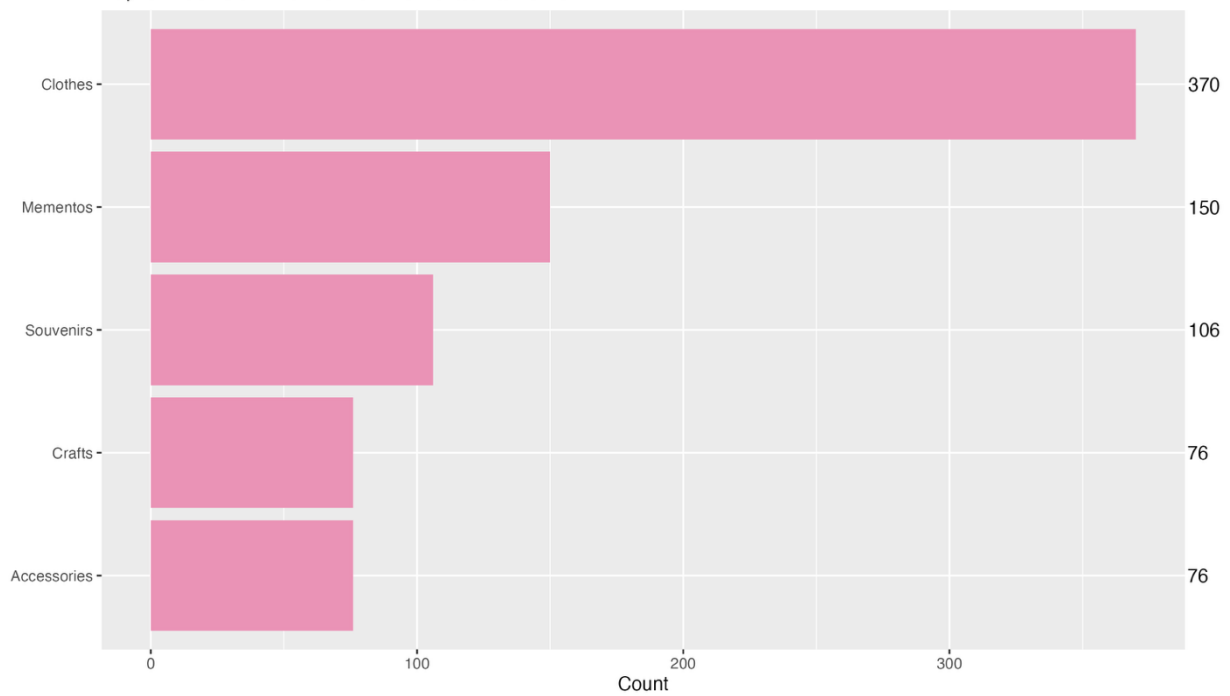
Top 5 Most Common Members/Converts/Friends Memorabilia Items



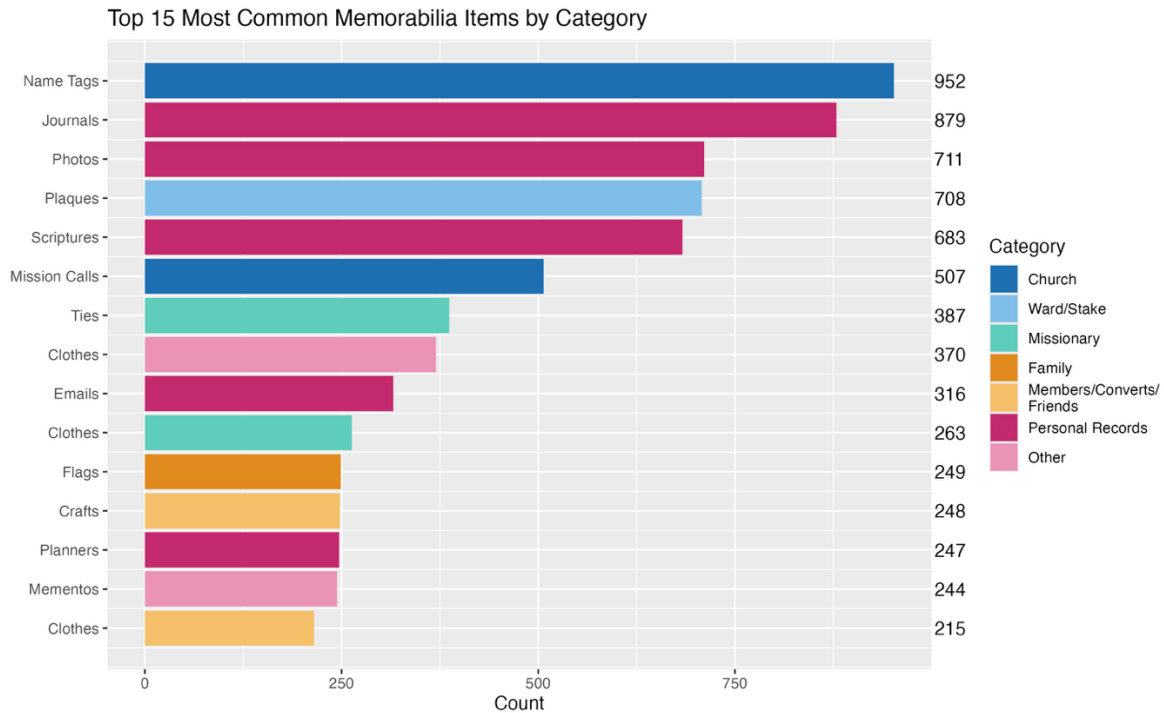
Top 5 Most Common Personal Records Memorabilia Items



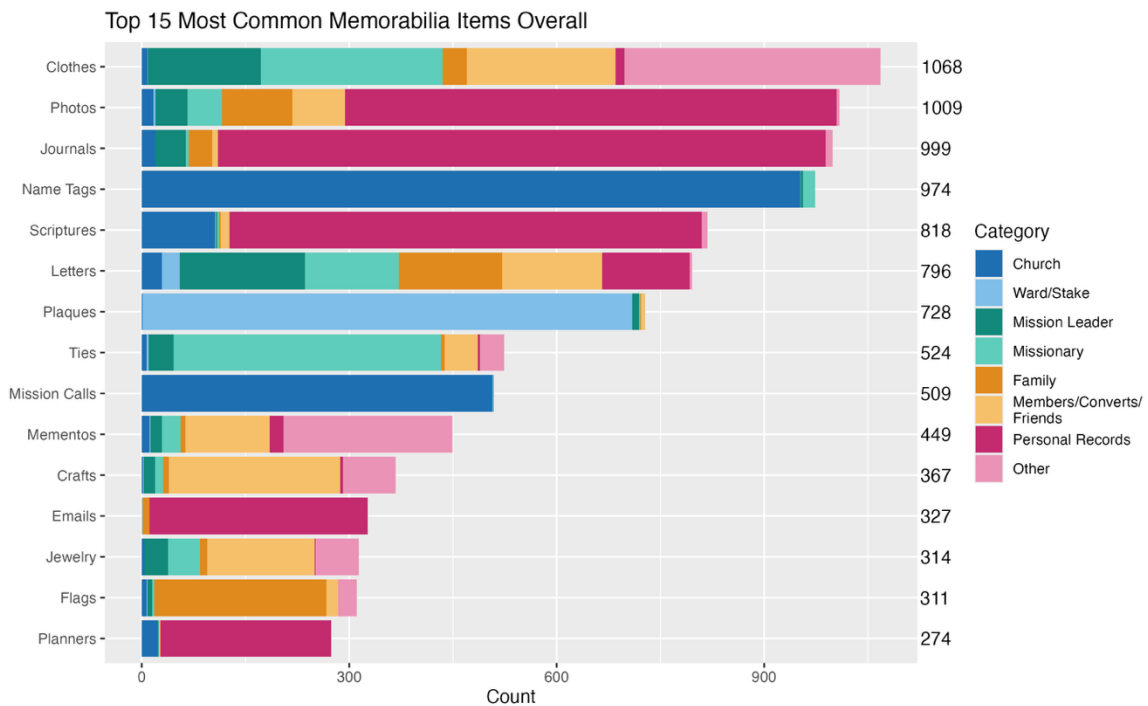
Top 5 Most Common Other Memorabilia Items



These graphs show the top 5 or 6 items reported in the memorabilia survey by category.

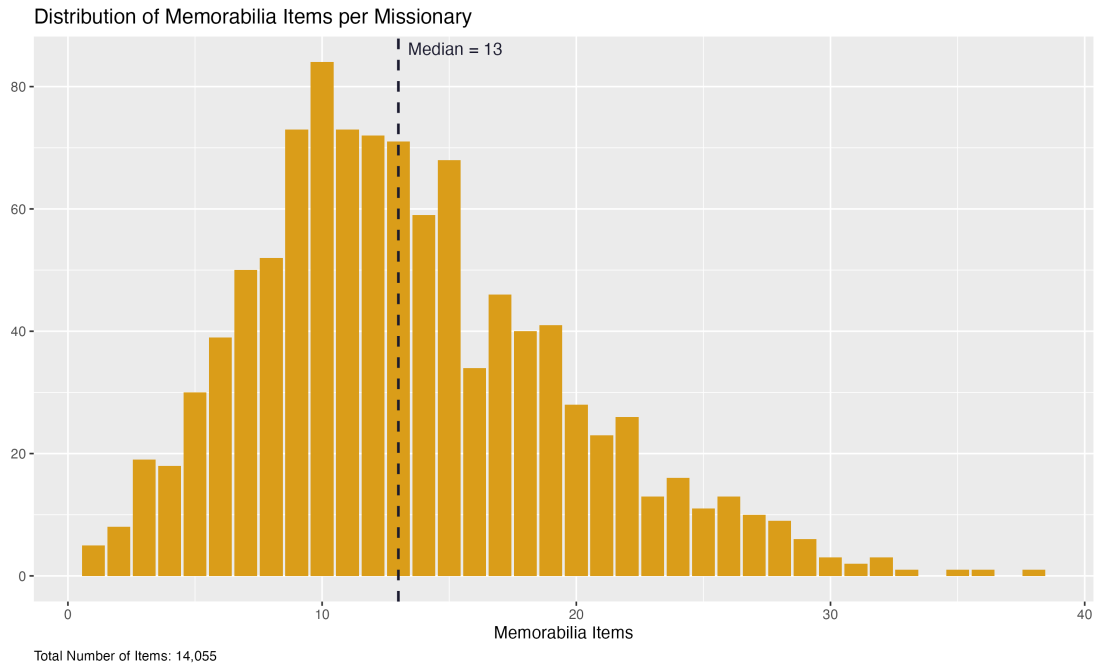


This shows the top 15 memorabilia items overall with category as the color for each bar. What stands out to me is the fact that the Records category accounts for the most items.

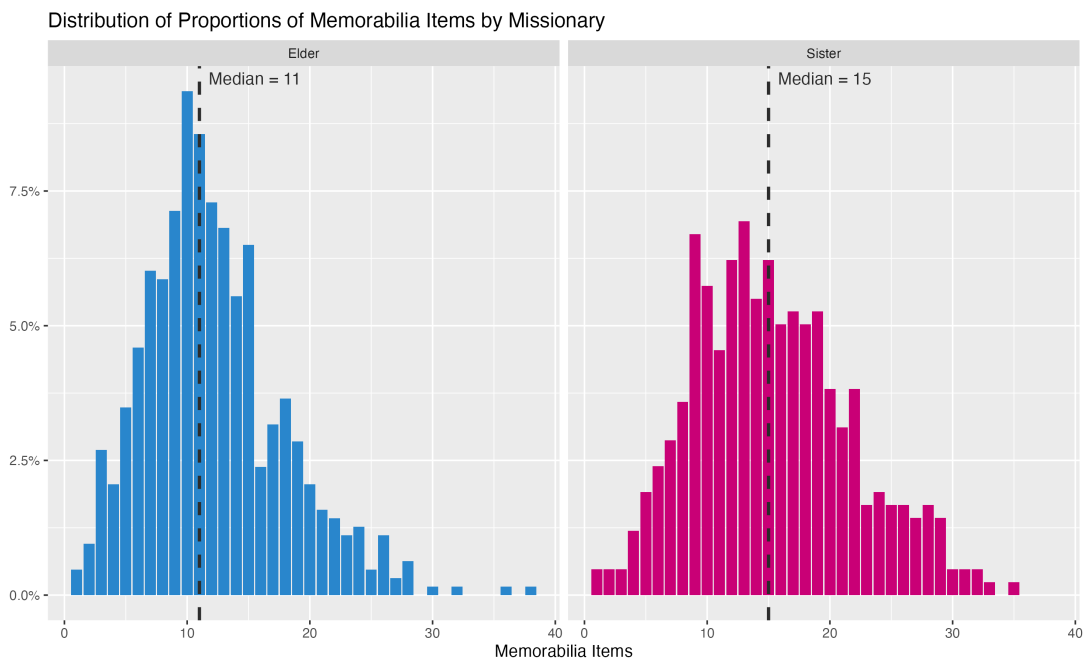


This shows the top 15 memorabilia items overall, regardless of which category it came from. I find it interesting that letters had come from so many different sources.

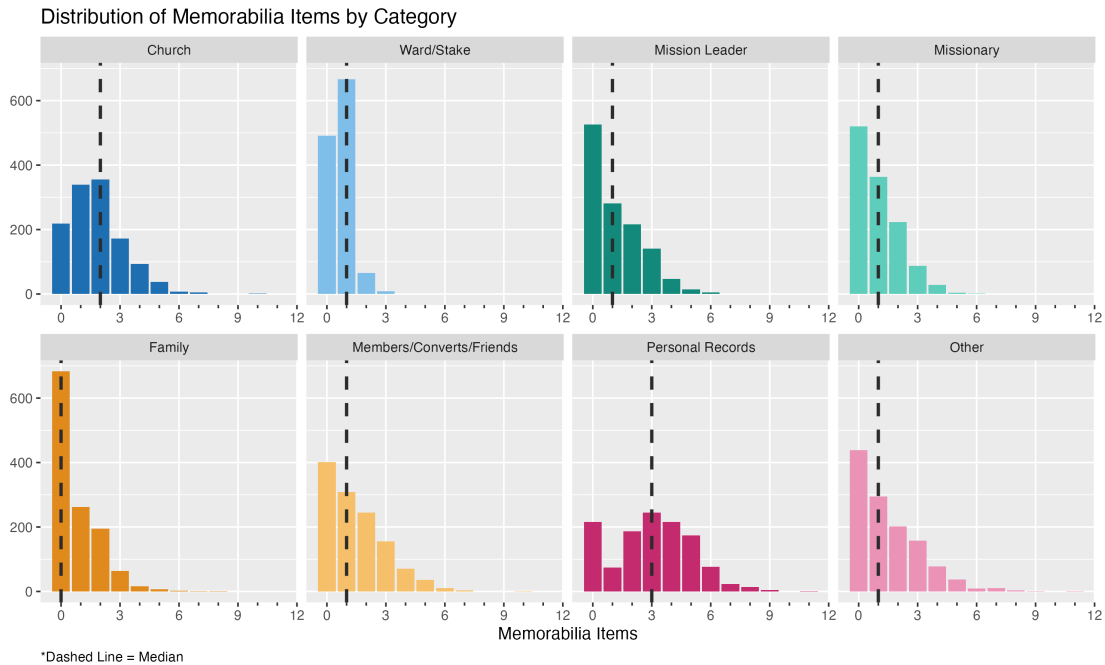
1.9 Number of Memorabilia Items



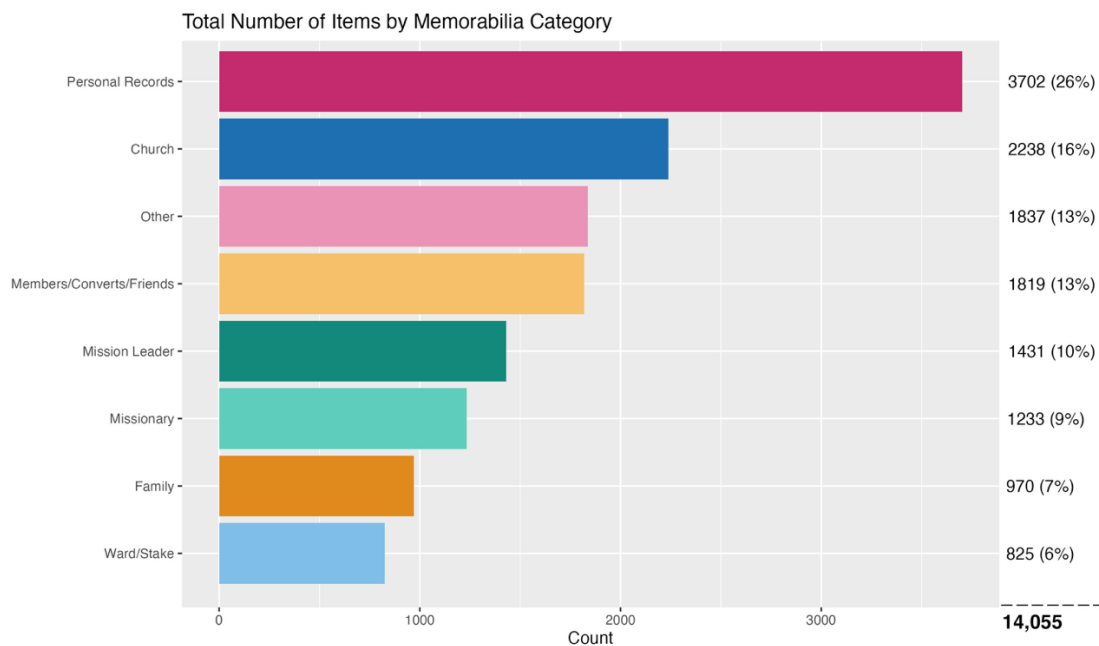
This graph shows the total number of memorabilia items mentioned. There is a right skew towards the higher number of items, with the median being at 13 items per missionary.



These graphs show the total number of memorabilia items mentioned by Elder or Sister missionary. We can see the Sisters are more evenly distributed, whereas the Elders are right skewed towards the higher number of items.

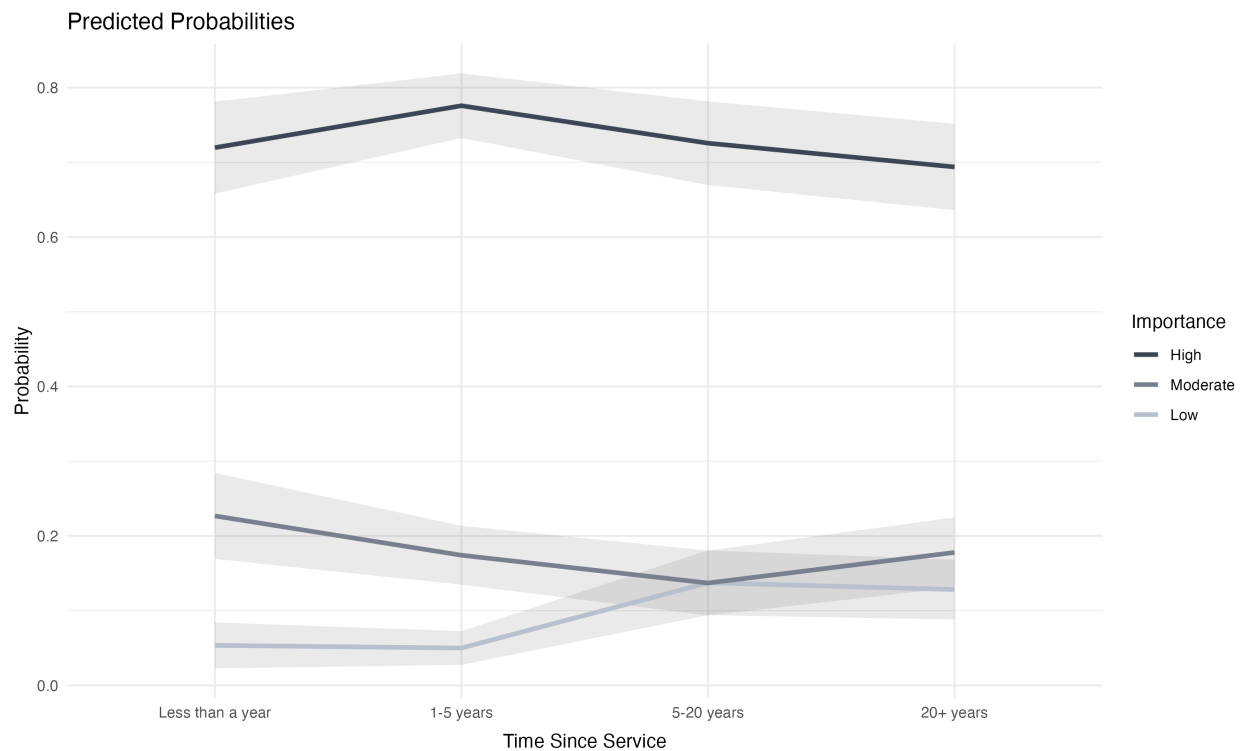


This graph shows the number of items reported by memorabilia category. The dotted line represents the median number of items. Most of the median items are at 2 items, with family having a median of no items, ward/stake as well as members/converts/friends having 1 item, with personal records having the most at 3 items.



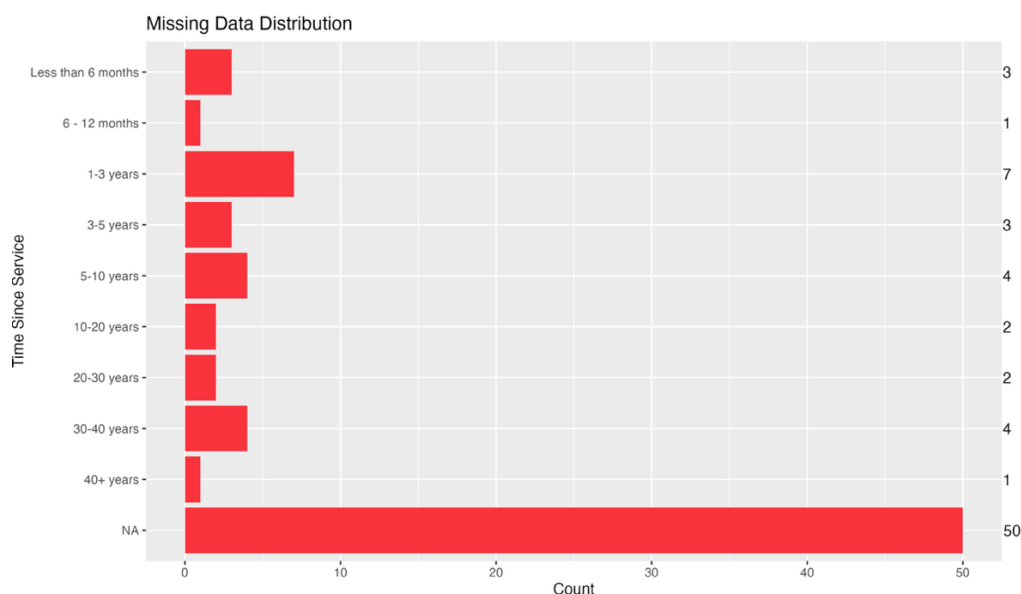
This shows the total number of items selected by memorabilia category. Personal records were by far the most common category report, with ward/stake items being the least common reported.

1.10 Ordinal Regression



This graph shows the predicted proportion of missionaries of how much they care for their memorabilia by how long it's been since they've served their mission. We can see that the two most recent groups, less than a year and 1-5 years, have the higher proportions of people who care about their memorabilia. They also have a decent amount of people who moderately care for their memorabilia, and hardly anyone who doesn't care for their memorabilia. The latter two groups, 5-20 years and 20+ years, have a large proportion who greatly care for their memorabilia. They also have a slightly smaller proportion of people who care for their memorabilia, with a greater proportion who don't care for their memorabilia.

1.11 Other Requests



This graph shows the number of people who didn't fill out the survey and how long it's been since they served their mission. Most people didn't get this far in the survey and fill it out, but for the few people who did fill it out, there isn't a clear trend of who wasn't filling it out. It appears that the distribution is somewhat similar to the overall distribution of people who filled out the entire survey.

Time from Missionary Service	1-5	6-10	11-15	16-20	21+	Total
Less than 6 months	0	14	26	23	13	76
6 - 12 months	9	30	44	29	21	133
1-3 years	11	50	75	52	36	224
3-5 years	10	39	41	25	13	128
5-10 years	13	57	57	31	26	184
10-20 years	1	17	16	10	12	56
20-30 years	10	27	21	11	15	84
30-40 years	3	14	15	8	6	46
40+ years	22	47	39	6	4	118
Total	79	295	334	195	146	1,049

This is a cross table of time since missionary service and number of items reported in the survey. The items are binned into multiples of 5 up to 20 where more than 20 items is its own category. There does seem to be a trend of more recently returned missionaries keeping more memorabilia as opposed to missionaries who've been back for a long time.